

CMP 459 Natural Language Processing (NLP)

Evaluation:

	Theory	Practical	Total
Sessional	50	-	50
Final	50	-	50
Total	100	-	100

Course Description

This course in Natural Language Processing is designed to introduce different levels of text analysis and computational aspects of written natural language. It will help students to appreciate complexities involved by focusing on a real world solution to each level of analysis.

Course Objectives

At the end of the course, students are expected to

1. define the nature and characteristics of a language and language processing;
2. apply computational techniques to model and analyze word, syntactic, semantic and pragmatic level of analysis;
3. conduct these processing techniques to design language supported systems.

Course Contents

- 1. Introduction to NLP** **3 hrs**
Origins and Importance of NLP
Challenges in NLP (Difficulties, Ambiguities and Evolution)
Language and Knowledge (Phonetics, Syntax, Semantics, Pragmatics and Discourse)
A multi-disciplinary field (Psychology, IR)
Application of NLP
- 2. Localization and Language Computing** **4 hrs**
Introduction to Localization
Key concepts of Localization
Unicode and Multilingual Computing
- 3. Finite State Machine (FSM) and Morphology** **6 hrs**
Introduction to FSM and FST
Morphological Processes
Principles of Word Construction (Suffix, Prefix, Stem, Affixes)
English, Nepali, Turkish
Morphological Representation and FSM
Lexicon, Morphotactics and Orthographic Rules
Morphological parsing and FST
Mealy Machine
FST Operations

4. Parts of Speech (PoS) Tagging and Hidden Markov Models (HMM)	4 hrs
PoS Tagsets Rule-based PoS tagging Stochastic PoS tagging Transformation based tagging	
5. Syntactic Analysis	7 hrs
Context Free Grammar(CFG) & Probabilistic CFG Words' Constituency (Phrase-Level , Sentence-Level) Parsing (Top-Down and Bottom-Up) CYK Parser Probabilistic Parsing	
6. Lexical Semantics	7 hrs
Lexeme, Lexicon, Senses, Lexical relations WordNet (Lexical Database) Word Sense Disambiguation (WSD) Word Similarity	
7. Pragmatics & Discourse Analysis	7 hrs
Monologue and Dialogue Reference Resolution Coherence and Cohesion Discourse Structure	
8. Application of NLP	7 hrs
Question Answering Machine Translation Sentiment Analysis Summary Generation	

Text Book

Jurafsky, Daniel, James H. Martin. 2000. *Speech and language processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*. Pearson Education: Singapore.

Reference Books

1. Manning, Christopher D., Hinrich Schütze. 1999. *Foundations of statistical natural language processing*. Cambridge: The MIT press.
2. PAN Localization - Guide to Localization of Open Source Software. Nepalinux Team, Madan Puraskar Pustakalaya, Nepal. Printworks, Pakistan n.d.
3. Unicode Demystified. A Practical Programmer's Guide to the Encoding Standard. Richard Gillam. Addison-Wesley, 2003.
4. Natural Language Processing with Python. Stephen Bird, Ewan Klein, and Edward Loper. O'Reilly Media, 2009. <http://www.nltk.org/book/>